Loan Application Risk Assistant

Build an AI-assisted loan application review service that summarizes applicant information, identifies missing documents, highlights potential risk factors, and generates an explainable recommendation for a loan officer.

Problem statement

A bank receives many loan applications with supporting documents such as income statements, employment details, credit notes, and policy guidelines. Loan officers spend significant time reading documents and checking eligibility rules. The organization wants an internal assistant that can help officers review applications faster while keeping the final decision under human control.

Expectations from candidate

Design a solution that accepts applicant details and document references.

Retrieve relevant policy clauses or eligibility rules before generating an answer.

Produce a structured output with summary, missing documents, risk indicators, recommendation, and confidence.

Include guardrails so the assistant does not make a final lending decision independently.

Store review history and a reviewer UI.

Add basic error handling for missing documents, invalid applicant data, and low-confidence responses.

Expected deliverables

Prompt or RAG pipeline design.

Sample policy documents or mock retrieval data.

Structured response schema.

Mandatory Technologies

The solution must use:

- Python
- LangGraph
- LangChain
- RAG
- Agent
- Guardrails
- Human-in-the-loop
- Local LLM such as Ollama
- Claude Code as the development assistant (optional)

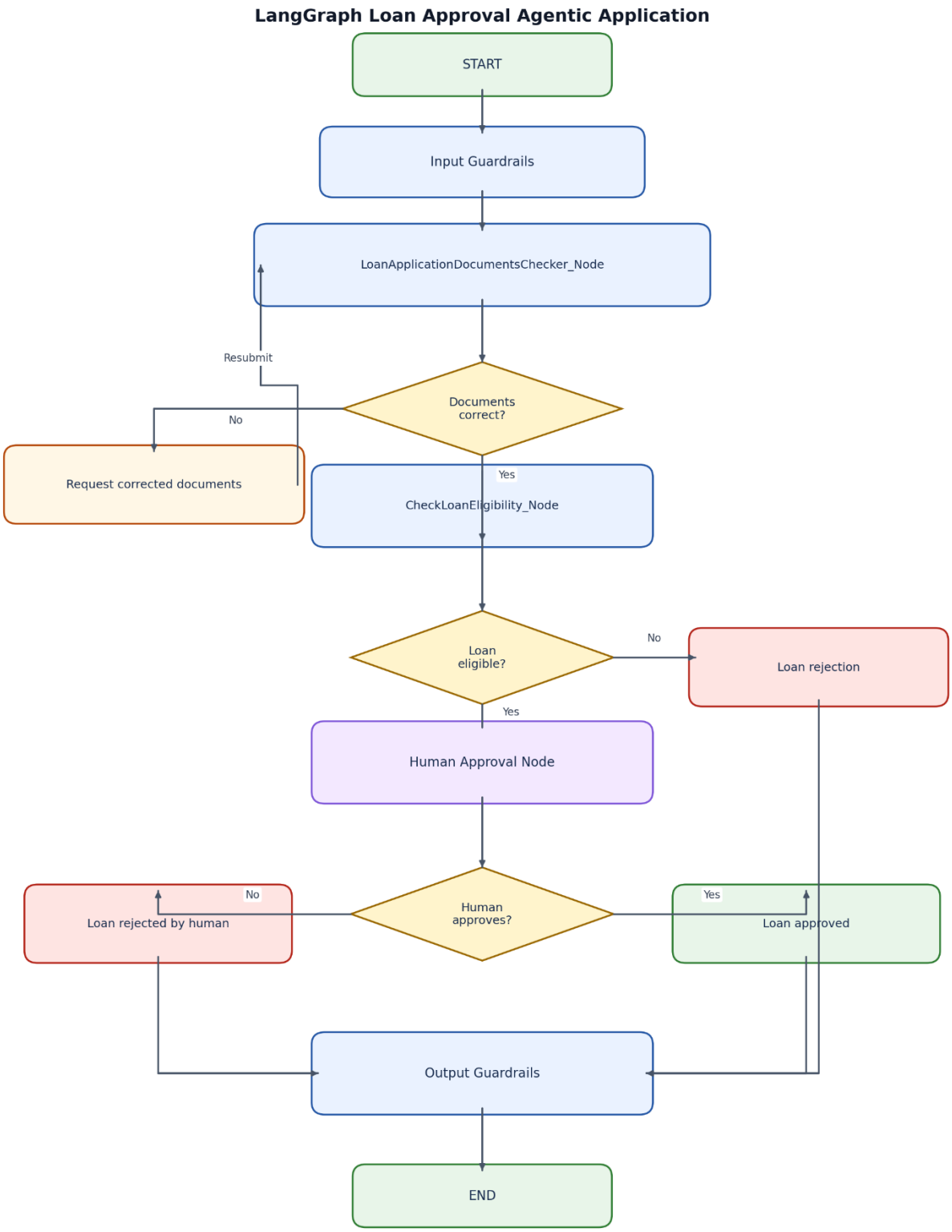
Functional Requirements

1. Knowledge Base

Create a small knowledge base containing documents related to:

- Loan eligibility criteria's
- Loan application required documents
- Loan interest rates and EMI details as per tenure

Implement RAG over these documents.


LangGraph Loan Approval Agentic Application
START
Input Guardrails
LoanApplicationDocumentsChecker_Node
Resubmit
Documents correct?
No
Request corrected documents
Yes
CheckLoanEligibility_Node
Loan eligible?
No
Loan rejection
Yes
Human Approval Node
Human approves?
No
Loan rejected by human
Yes
Loan approved
Output Guardrails
END

Note:

1. Once you complete the use case, simply copy source code of all python files into email body in plain text format & send the email to me anand.kulkarni1@zensar.com.
2. There is no score given to this use case. Thus, the use case submission is optional but highly recommended.